Rockchip RKAI Developer Guide

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Trademark Statement

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Preface

Overview

This document presents the steps and guides for rapid verification and development of large language models and visual large models.

Product Version

SoC	Kernel Version
RV1126B	Kernel 6.1

Target Audience

This document (this guide) is mainly intended for:

Technical support engineers

Software development engineers

Revision Record

Version	Author	Date	Modification Description
V1.0.0	Cherry Chen	2025-09-30	Initial Version

Contents

1. Introduction

The RKAI module is a rapid verification module and adaptation example for SDK integration with Large Language Models (LLM) and Vision Language Models (VLM). The source code of this module is open-source and can be found at the path: `app/rkai`. The directory structure is as follows:

```
tree  rkai -L 1
rkai
├── app_conf #Configuration files and model files
├── CMakeLists.txt
├── common  #General code
├── LICENSE
├── README.md
├── rkai_LLM  # LLM model API source code
├── rkai_VLM  # VLM model API source code
└── samples   # Adaptation use cases

5 directories, 3 files
```

- app_conf is the general configuration path, which includes three parts: model files, model configuration files, and model input files. This part can be directly integrated into the board for use.
- common contains general code for the RKAI module, such as `CNPY`, etc.;
- rkai_LLM is the general API for LLM deployment;
- rkai_VLM is the general API for VLM deployment;
- samples provide verification cases for LLM/Vision/VLM.

2. Quick Verification

The SDK comes with the RKAI module pre-compiled by default and integrates the `Qwen3-0.6B` model, allowing users to quickly verify the process of large models.

2.1 Quick Verification of LLM

The LLM adaptation program is `rkai_LLM_Demo`, with the following Usage:

```
Usage: rkai_LLM_Demo [options]
Options:
  --llm_model <path>   RKLLM model file
  --config    <path>   Configuration ini file
Example:
  rkai_LLM_Demo --llm_model /usr/share/app_config/models/Qwen3-
0.6B_W4A16_RV1126B.rkllm   --config rkai_config.ini
```

The default inference command is:

```
rkai_LLM_Demo --config rkai_config.ini
```

The program comes with 4 pre-set questions, from which users can select by entering numbers, or input questions directly through the terminal.

```
*********************Enter the corresponding number for the following questions
to get answers/ or input custom questions*******************
[0] Translate the following modern text into classical Chinese: When the spring
breeze is gentle and the sun is bright...
[1] Write an ancient poem with the theme of "Ode to Plum Blossoms", including
elements such as plum blossoms and snow.
[2] Upper couplet: Used to watch thousands of sails pass by the riverbank
[3] Translate this sentence into Chinese: Knowledge can be acquired from many
sources...
[4] Translate this sentence into English: RK3588是新一代高端处理器...

*************************************************************************
```

The model output consists of two parts: 1. Model response; 2. Performance.

You can also convert the model yourself and use `rkai_LLM_Demo` for model inference verification. Detailed instructions are in the third section.

2.2 VLM Rapid Verification

VLM actually refers to Vision and LLM, which can be verified separately or together. Due to the large memory footprint of the models, it is not convenient to integrate them into the SDK. Therefore, users can quickly verify the VLM model by downloading from the following URLs.

The Vision model download link is: Vision Model;

The LLM download link is: LLM Model.

Push the downloaded models to the board's `app_config/models` directory. Then, you can use `raki_VLM_Demo` for rapid verification. `Usage` is as follows:

```
Usage: rkai_VLM_Demo [options]
Options:
  --vision_model <path>   Vision RKNN model file
  --llm_model    <path>   LLM RKLLM model file
  --image_path   <path>   Input image file
  --config       <path>   Configuration ini file
Example:
  ./rkai_VLM_Demo --vision_model fastvlm-0.5b-fp16.rv1126b.rknn --llm_model
fastvlm-0.5B-w8a8_level0.rv1126b.rkllm --image_path image.jpg --config
rkai_config.ini
```

If the downloaded models are pushed to the `app_conf/models` path, then you can use the following command for inference:

```
rkai_VLM_Demo --config rkai_config.ini
```

If the downloaded models are pushed to other paths, then you need to manually specify:

```
rkai_VLM_Demo --vision_model xxx.rknn --llm_model xxx.rkllm --config rkai_cfg.ini
```

The model output consists of two parts: 1. Model response; 2. Performance.

3. Custom Models

Users can convert models on their own and perform inference using `rkai_LLM_Demo` and `rkai_VLM_Demo`. The model conversion tool download link is: https://github.com/airockchip/rknn-llm. If users customize their models, then `rkai_cfg.ini` needs to be modified accordingly.

1. In `rkai_cfg.ini`, the [llm_cfg] section contains LLM parameters, including model hyperparameters that users can adjust. The parameters are as follows:

Parameter	Default Value	Description
llm_model	models/Qwen3-0.6B_W4A16_RV1126B.rkllm	Can be defined externally with `--llm_model`
max_context_len	1024	Maximum context length
max_new_tokens	128	Maximum output tokens
skip_special_token	1	Whether to skip special characters
base_domain_id	0	Memory usage method, default is 0
top_k	1	Less than or equal to 1

2. In `rkai_cfg.ini`, the [vlm_cfg] section contains VLM parameters, including model hyperparameters that users can adjust. The parameters are as follows:

Parameter	Default Value	Description
vision_model	models/fastvlm-0.5b-fp16.rv1126b.rknn	Can also be defined externally with `--vision_model`
llm_model	models/fastvlm-0.5B-w8a8_level0.rv1126b.rkllm	Can be defined externally with `--llm_model`
image_path	inputs/image_1024x1204.npy	Supports input jpg images, specified with `–image_path`
max_context_len	512	Maximum context length
skip_special_token	1	Whether to skip special characters
base_domain_id	0	Memory usage method, default is 0
top_k	1	Less than or equal to 1

3. In `rkai_cfg.ini`, the [prompt] section is the system prompt, and users need to modify this part when customizing their models. The parameters are as follows:

Parameter	Default Value	Description
system_prompt	<|im_start|>system\nYou are a helpful assistant.<|im_end|>\n	Default system prompt for qwen3
prompt_prefix	<|im_start|>user\n	Can be obtained from the model's config file
prompt_postfix	<|im_end|>\n<|im_start|>assistant\n	Can be obtained from the model's config file

4. In `rkai_cfg.ini`, the [tensor] section is the image placeholder for the VLM model, and users need to modify this part when customizing their models. The parameters are as follows:

Parameter	Default Value	Description
img_start	<|im_start|>	Image placeholder, can be obtained in the model config
img_end	<|im_end|>	Image placeholder, can be obtained in the model config
img_content	<|im_content|>	Image placeholder, can be obtained in the model config

4. Detailed API Documentation

Users can integrate applications by calling APIs themselves. The APIs are fully open-source, and the following will introduce the APIs.

4.1 LLM API

The APIs for initializing and deinitializing the LLM model are as follows:

API	rkai_llm_init
Description	LLM Initialization
Parameters	rkaiCtxLLM* ctx: LLM handle
	rkaiCfg* cfg: LLM parameter configuration

API	rkai_llm_release
Description	LLM Deinitialization

The APIs for LLM inference deployment are as follows:

API	rkai_llm_run
Description	LLM Inference
Parameters	rkaiCtxLLM* ctx: LLM handle
	rkaiCfg* cfg: LLM parameter configuration

4.2 VLM API

VLM's Vision Model API:

API	rkai_vlm_vision_init
Description	Vision Initialization
Parameters	const char *model_path: Vision model path, in .rknn format
	rkaiCfg* cfg : LLM parameter configuration

API	rkai_vlm_vision_release
Description	Vision De-initialization

APIs for Vision Inference are as follows:

API	rkai_vlm_vision_run
Description	LLM Inference
Parameters	rkaiCtxVision *rkai_ctx: Handle
	void *img_data: Input image buffer, in rgb888 format
	rkaiVisionToLLM *vtol: Parameters and buffer for vision to llm

API	dump_tensor_attr
Description	Dump basic information of the model
Parameters	rknn_tensor_attr *attr: Model tensor information
	int32_t is_inputs: Whether to dump inputs

VLM's LLM API:

API	rkai_vlm_llm_init
Description	LLM Initialization
Parameters	rkaiCtxLLM* ctx: LLM handle
	rkaiCfg* cfg :LLM parameter configuration

API	rkai_vlm_llm_release
Description	LLM De-initialization

APIs for LLM Inference Deployment are as follows:

API	rkai_vlm_llm_run
Description	LLM Inference
Parameters	rkaiCtxLLM* ctx: LLM handle
	rkaiCfg* cfg :LLM parameter configuration
	rkaiVisionToLLM *vtl: Receive information from Vison